

ActInf GuestStream 118.1 ~ Tadahiro Taniguchi "Co-Creative Learning via

GuestStream | Tadahiro Taniguchi | 1:10:59

Hello, welcome everyone.

It is August 20th at the stroke of zero UTC on the 21st in 2025 and it's Active Inference

Guest Stream 118.1.

We're here with Tadahiro Taniguchi and we'll see a presentation on co-creative learning via Metropolis Hastings' interaction between humans and AI.

And then I will read any questions in the live chat and ask some things about this fascinating research.

So thank you for joining again, Tadahiro, and to you for the presentation.

Okay, thank you very much for inviting me to this great guest swim again.

So I'm Tadahiro Taniguchi from Kyoto University.

And today I'll be talking about co-creative learning via Metropolis Hastings' interaction between humans and AI.

And this is based on our preprint, which is recently available on archive.

So this is the co-creative learning.

It's a new theory of human-in-the-loop machine learning method.

So I'll also discuss collective predictive coding hypothesis, which I talked in the previous guest swim.

Maybe you can check it later.

But as a background of this work.

Okay, let me start.

So here is an overview of our main contributions of this paper.

The first, we formally propose co-creative learning.

This is a novel paradigm where humans and AI collaboratively achieve symbol emergence.

In other words, the formation of the external representation

by mutually integrating their partial perceptual information to build shared representation.

So this fundamentally shifts the perspective of human-AI alignment or teaching process, moving beyond traditional unilateral approach.

The second, we introduce an experimental framework based on the Metropolis Hastings naming game.

Through this framework, we propose the comparing empirical evidence,

demonstrating that human-AI pairs can effectively engage in this co-creative learning process.

So the visual of this slide illustrates the idea of mutual interaction between the human and AI centering around the shared world.

Okay, so let's consider why co-creative learning is so crucial currently in this phase.

So the first one is limitations of unilateral alignment or teaching.

Traditionally, unilateral alignment framework or teaching,

typically supervised learning based on ground truth levels provided by human annotators, have significant limitations.

For instance, AI systems already excel beyond average human performance in many areas.

the possessing distinct and often superior domain knowledge.

Furthermore, the human and AI inherently have heterogeneous knowledge.

Our ways of understanding and processing information are fundamentally different.

So when, and also when embodied AI agents start exploring the world autonomously,

it becomes questionable whether we should enforce our existing human knowledge and values as a sole ground truth.

So we should respect, in some sense, we should respect AI's perception and or knowledge to bring our knowledge better status.

Okay, another thing is the bidirectional, the influence.

So actually, the large language models, for example, actively impact the human language use and decision making, as shown by recent studies.

This, the mutual influence clearly highlights the limitation of the perspective of purely one-side alignment strategy, teaching strategy.

Therefore, looking beyond the current generative AI error,

it is important that we establish a new system theory,

the principle, and the learning method

to achieve true human AI symbiosis,

the symbiosis, considering bidirectional alignment or teaching.

That can be called co-creative learning process.

So, to position co-creative learning more intuitively,

from the machine learning perspective,

let's briefly and roughly review existing traditional unilateral machine learning frameworks.

So, in the most common,

the unsupervised or the reinforcement learning,

the human's perspective,

the human's perspective can be written like that.

The XN human is the human's observation,

and SN is the kind of cross-label or description

of the human's perception of information.

So, we form the inner society,

we form the relationship between the labels

and the particular information, in some sense.

So, and this is effectively serves as the ground truth

or the target or reward,

the source of reward for the AI to learn from.

The AI's understanding is therefore fundamentally bounded by the human's perspective.

So, in contrast, in unsupervised or self-supervised learning, the AI agents learn in isolation, inferring knowledge solely from its own observations.

In this case, in this case, its understanding is bounded only by the limits of its own perceptual data.

So, a key takeaway from both these traditional approaches is that the only partial information, either human or AI, is used for the learning process.

So, this highlights a critical limitation.

The NISA approach fully liberates the combined, potentially richer, the information available when humans and AI interact.

So, this brings us to our proposal, the new bilateral framework, co-creative learning.

Here, the goal is not for one agent to teach the other, but for both to engage in bidirectional message passing through language games and communication to integrate their partial observations.

So, instead of trading one agent's perspective as a sole ground truth, the learning target for the dyad that becomes the true posterior distribution conditioned on the joint observation of both agents.

Here, X and AI and X , the human.

This is crucial distinction.

So, by integrating their distinct and partial information,

this approach allows
emergent shared understanding
to surpass the knowledge
and perception
that either the humans
or the AI could have
have formed along.

The diagram on this slide
illustrates this mutual interaction
and the formation of a joint
more complete understanding.

So,
and, but, you know,
how can we do this?

That is a question.

And before diving into

today's main topic,

let me briefly introduce

or review

the collective predictive coding
hypothesis,

which I proposed last year.

And we gave a talk

in this guest stream last time.

So,

this hypothesis extends idea
of well-known predictive coding
to social and linguistic phenomena.

So,

it states that

symbol systems,

including language,

are formed

through collective

predictive coding by humans.

In other words,

a human society,

which can be regarded

as symbol emergence system,
the systems performing symbol emergence,
that technically,
can be understood
as a social representation
learning system.

As a result,
the information of the world
may be encoded
in the distributional semantics
of our language.

The hypothesis also suggests
that this is why
large language models
appear to understand the world.

They inherit
this structure
through vast amount
of distributional,
distributed human generated text.

By the way,
the diagram illustrates
this complex process
where
the emergent,
where the individual
inference
and action loop
contribute to an emergent,
the symbol system
that in turn
constrains
the organized
collective behavior.

So,
let me illustrate.

So,
maybe the

explanation was too quick.

So,

let me illustrate

the CPC hypothesis,

collective predictive coding

process,

the bridging the idea

of emergent communication

and predictive coding

or free energy principle.

First,

let's start from

the predictive coding

or word model learning.

So,

there is a person,

the internally,

the he,

in turn,

he or she

internally has

internal representation

over world models.

And we are

living

with

action perception

loop.

So,

this is very roughly

speaking

the inference process.

But we humans,

we are so many

and through communication

we form language.

So,

following

the predictive
coding hypothesis
through language game
we can form
language-like
social,
the external
representations
through this
learning,
total learning
process.

I think
through this process
we model
and adapt
to the world
collectively.

So,
but here
let me take
another view.

Until now,
we believe,
we assume
that each
person
is an independent
and autonomous
agent.

Let's change
and take
another viewpoint.

First,
language is here.
Language is a kind
of autonomous
system.

And
they have sensors
and also actuators.
That's us.
The language uses,
utilizes us
to model
the world
through interaction.
From this
viewpoint,
we are subordinate
to the language
system.
So,
this kind of view
is often mentioned
in sociology
or other
social science
area
or cultural
psychology.
So,
I think
this kind
of dual
viewpoint
is very
interesting
to extend
our view
of cognition
to the
social
viewpoint.
And actually,
this dual

viewpoint
actually
gave us
opportunity,
gives us
opportunity
to formalize
this process
from the
viewpoint
of free
energy
principle.

So,
based on
the generative
model,
representing
CPC,
I mean,
this kind
of the
language model.

Let's consider
language
is a kind
of brain
having these
sensor motor
systems.

So,
the following
the
graphical model,
the
pro-bisic
generative model,
the describing

such kind
of process,
we can define
the free
energy
in the same
way
as
the normal,
usual,
the free
energy
principle.
Interestingly,
we can
naturally
derive
this kind
of
formulation,
and interestingly,
the final
form
can be
divided
into two
parts.
of the

So,
the first one
is an
ordinary
variation of
free energy
term.

This is the
sum of the

variation of
free energy
of a number
of agents.
in addition to
that,
an interesting
term pops
up.

We call
this
collective
regularization
term.

This term
tries to align
agents'
external
representations,
 W ,
 W ,
and
with each
agent's
internal
representation,
 Z .

With this
term,
the problem
of
minimization
individual
free energy
becomes
the
minimization
of

free energy
of a
single
total
system.

I think
this book
can extend
the free
energy
principle
from the
individual
to the
collective
level
with a
clear
relationship
to
the
emergence
of
symbol
systems
or
language.

Okay.

So,
as a
simplest
instance
of
the
theory
of
collective
predicted

coding,
let me
introduce
the
Metropolis
Hastings
naming
game,
which we
previously
proposed.
It is a
type of
language
game
built on
the
solid
mathematical
foundations.
This will
serve as a
basis
of
co-creative
learning.
I will
introduce
Delta.
Okay.
How does it
work?
So,
here's the
outline.
First,
in the
perception

phase,
the
speaker
and
listener,
observe
the
same
object
and
infer
their
internal
representations.

We assume
joint attention
here,
which is
crucial for
human language
acquisition.

Next,
in communication
phase,
the speaker
provisically
utter a name
for the
object.

The listener
then determine
if it accepts
the naming
or not
with a certain
probability.

The probability
depends on

its current
belief state.

Specifically,
this probability
show.

Probability
shown in the
equation
on the
slide
reflects how
well the
listener's
belief matches
the name
proposed by
the speaker.

So,
if the listener
agent does not
find the
incoming sign
believable,
it can reject
the proposed
name.

This ability
to reject
is crucial.

Third,
the learning
phase.

After the
communication,
the listener
update
its internal
parameter

related to
representation,
learning,
and naming.

Finally,
the turn-taking
step,
the speaker
and listener
alternate
as their role
and the game
continues.

Overall,
this is a
very very
simple game.

But importantly,
this game
has a solid
mathematical basis.

So,
let's dive into
the data.

If the
acceptance
decision
is
made
according
to this
probability,
we can
mathematically
prove
that the
naming game
is equivalent

to the
metropolis
haying
algorithm
for
Bayesian
inference
or
the
shared
representation
shown
in
this figure
on the
left.
So,
let's begin
with this
graphical
model.
So,
here,
OA
and
OB
or
we can
rename
this
XA
or XB
that represents
a visual
observation
of each
agent.
They are

the agents
with the
visual input
here,
but any
kind of input
is okay,
basically.
Z is,
ZA
and ZB
are the
internal
presentations,
latent
variables.
We also
assume
the existing
of a
variable
W,
which serves
as a
prior
for these
two
observations,
two
internal
variables.
So,
if we
treat
W
as a
latent
variable,

this graphical
model becomes
very similar
to the
multi-modal
or multi-view
variational
auto-encoder
or latent
variable
models,
and if the
two agent's
brains were
directly connected
and could be,
could use
both observations
simultaneously,
yes,
this variable
can be
inferred
in the
natural way
and would
encode
the combined
information
from both
agents'
input.
However,
in reality,
they are
independent
cognitive
systems.

So,
the algorithmic,
so,
let's introduce
a trick.
Algorithmically,
we can decompose
the model
into two parts.

So,
left-hand side
is agent
A,
and right-hand side
is agent

B.
And we,
let's consider
the kind of
alternative,
the inference,
the method
for this

W.
Now,
think about
the sampling
process.

Agent A
proposes
a candidate
of latent
variable
WD
by sampling
from its
posterior
distribution.

And
the
proposed value
is passed
to agent
B.
The agent
B can
decide
whether
to accept
the sample
from
a sample
based on
certain probability.

So,
this is,
this process
is,
this process
can be
designed
by using
the
Metropolis-Hastings
algorithm,
which is
a kind of
Marco
J.
Monte Carlo
MCMC.
So,
actually,
under certain
conditions,
this guarantee

is that
shared signs
are formed
and
that
categories
emerge
among
the agent
to approximate
the posterior
distribution
based on
the agent's
observations.

So,
and importantly,
this,
what I
explained here
is
completely
equivalent
to this
naming game.

So,
it is
proved that
this naming game
itself
is
Bayesian
inference,
approximate Bayesian inference
procedure.

So,
this is a very
important point.

This is a very
important point.
So,
in the previous
paper,
we tested
this idea
on some
dataset
and we
showed that
the agent,
the two
agent
playing
Metropolis-Hastings
naming game
can effectively
integrate
their
perceptual
information
and form
and share
signs
and signs
I mean
the categories
across
the agent
and actually
such kind
of process
even
outperform
the
learning result
achieved

by a single
agent.
So,
without any
ground truth,
this kind of
method-point-hasting
interaction
can improve
the
representation
learning
process
in some
sense
the
knowledge
the formation
process.

Okay,
so this is the
background
and actually
after
this
computation
study
you know,

the
we
the
as
you see,
as you saw
in my talk,
I always

assume that
this is
something happening
that in
now society,
human.

So,
to show that,
we need to check,
we need to validate
that if
this kind of
things
is happening
in human
interaction.

So,
we designed an experiment
to test whether
human
participant
when playing a game
similar to the
M.H.

Metropolis-Haysing's
naming game
would follow
the acceptance
probability
calculated
by the
algorithm.

The results
as shown
in the
graphs
on the
right

are
positive.
We found
that
human
acceptance
behavior
indeed
are mostly
aligned
with
theoretically
derived
M.H.
acceptance
probability.
This is
crucial
finding
because
it suggests
that
humans
to a certain
extent
follow
similar
patterns
to the
Metropolis-Hastings's
naming game
providing
empirical
support
for the
idea
that
human

behavior
on
simple
emergence
might
follow
a CPC
mechanism.

This
paves
the way
for our
recent
work
integrating
AI
and
human
into
this
framework.

So,
this is
the beginning
of
actually
today's
main
topic.

So,
a simple
way to
introduce
co-creative
learning
is to
view
it

through
the
lens
of
a
metropolis-haysing
style
interaction
between
human
and
AI.

Like this.

So,
okay,
let's jump
into
this section.

So,
given this insight,
we then move to
the core
of our current
work,
the
realisation
of
collective
predictive
coding
within
a
human
and AI
collaborative
system.
As depicted
in this

graphical model,
we envision
the system
where a
human
and AI
agent,
each with
partial observations,
engage in
bilateral
message passing
through language
game or
communication.

This interaction
facilitates the formation
and influence of shared representation.

And represented by the Q , S , N .

this is the kind of sign and shared latent variable.

And X , A , X , AI and X , human.

This means that AI's observations and human's observations.

Okay.

This is what we define as co-creative learning.

The key idea here is that instead of one agent simply teaching the other,
the both mutually integrated their partial perceptual information.

The display process enabled by the mechanism like the MHNG, Metropolis-Haysynx naming game,
allows them to jointly construct a richer shared external representation that neither could achieve alone.

This is a direct instance,

uh,

uh,

uh,

instantiation

of the CPC hypothesis

in a human AI triad.

So more formally,

to be prepared,

we formally define co-creative learning

as an interactive decentralized
major inference process.

As shown in,
in definition 1,
uh, in our,
in our,
in our playprint.

Through message-based update,
like the Metropolis-Haysynx naming game,
like the Metropolis-Haysynx naming game,
jointly construct a local belief distribution.

The crucial aspect is that this process ensures the expected
of the collective free energy,

FT,
decreases over interaction steps.

This means that the Amarco-Chain formed by their interactions
converges to the true posterior distribution,
conditioned on a joint observation of both AI and the humans.

So what's the remarkable about this mechanism is that it allows for
improvement of Monte Carlo estimates of the evidence P ,
 X_{AI} ,
 X ,
human.

Without requiring either agent to disclose their private observation or internal gradient.

So this decentralized nature is a key advantage in enabling robust and private knowledge co-construction.
Okay.

So with this formal definition in mind,
our primary objective for this experiment was to empirically test this concept.

We conducted an experiment where joint attention naming game,
this is a kind of, you know,

the experiment,
experiment with participants,
asking to play the MH naming game like,
you know, naming game.

So this game was played between human participant and the computer agent.

The central question we aim to answer was,
can the human and the computer effectively integrate their respective partial observations
through this naming game?

So thereby,
they engage in the decentralized Bayesian influence,
or as we term it, co-creative learning.
To ensure this,
each agent was given only a partial missing subset of the full observational information.
As you can see in this diagram,
the human might have certain dimensions of features.
while the agent has others,
and some are missing for each.
So this setup makes accurate categorization impossible,
based on solely on individual information.
Thus,
the human might have certain information integration
through the communication.
We wanted to see if they could
collaborateally fill in these missing pieces
and reach shared understanding.
So let's detail our material methods.
The stimuli consisted of images depicting circles with a radial notch.
So these were generated based on five-dimensional vectors,
sampled from one of three ground truth Gaussian distribution.
And using the CIE-lab color space,
which is perceptually uniform.
So the parameters were chosen.
So that categories would overlap significantly in the partial observation space.
Like this.
But crucially, as shown in figure,
this part,
humans and computer agents receive different partial observations.
human participants view a grayscale rendering based on the original figure.
And the agent only can see the color information.
This partial observability makes accurate categorization impossible,
based solely on individual information.
So making communication essential.
So that DIA's task was to collaborateally categorize these stimuli
into up to three categories.
Yes, the ABC.
By playing the naming game,

this side, figure B,
illustrates the experimental procedure flow
involving initial categorization,
initial categorization,
and the naming game,
turns and role switching.
So this is actually very difficult task for human participants,
human participants,
human participants,
as you can imagine.

Okay.

This is a very, this is a process.

The, the slide shows a screenshot of the experimental interface
that used in our experiment.

First, participants were asked to categorize the given images by dragging
then into rappelled boxes, as seen on the left.

During the naming game,
during the naming game,
one image is presented on the right.

On the right.

And the naming agent propose a name or sign for the image by clicking the button.

The listener then decides whether to accept the proposed proposed name by clicking and accept or reject.

Sorry, the instruction is in Japanese.

After this decision, the listener is prompted to revise their own categorization if needed.

This intuitive interface supports the partial observation and sign exchange between the human and computer
agent,

facilitating the interactive process.

Okay.

So the experiment was conducted online by a web browser.

We recruited 90 participants through CloudWorks,
Cloud Sourcing Service,

and after filtering for engagement and adherence to instruction,
69 participants were included in our final analysis.

The experiment typically lasted about 90 to 60,
to 120 minutes,

like 2 hours.

The participants received compensation for their time.

Okay.

So for comparison,
participants were divided into 3 distinct groups.
So each interacting with computer agent,
employing a different acceptance strategy for human proposal.
Group one, the MH group, the Metropolis Hastings, engaged with an AI that accepted the proposals based on the Metropolis Hastings acceptance probability,
serving as our primary co-creative learning condition.
Group two, the Always Accept group.
The Interactive with an AI that always accepts proposals,
acting as a proxy for traditional supervised learning.
The human suggestion is always true.
You know, the human suggestion is always true.
You know, the human suggestion is always true.
Finally,
Group three,
the Always Reject group.
The Always Reject group.
The communicated with an AI that always reject human proposals,
the mirroring and unsupervised learning scenario.
So the AI just learned the form categories from their observation.
These conditions allow us to systematically evaluate the effectiveness of the MHNG in facilitating co-creative learning.
So, let's look at the result for clustering accuracy measured by adjusted land index, AI.
Table one summarizes the initial and final mean AI score for both human and computer agents across all groups.
A key finding is that participants in MH group achieve the highest categorization accuracy and sign agreement, particularly on the AI side.
When compared to both the Always Accept and Always Reject group,
the computer agent in MH condition,
the computer agent in MH condition achieved a significantly higher final mean AI.
This clearly indicates that the MH-based interaction most effectively facilitated shared understanding between human and AI,
and AI engaging a more collaborative and accurate modeling of the world.
So, while interacting generally helped the human improve,
the MH mechanism was uniquely effective for the AI agent to leverage the interaction for super-spiral performance.
Okay, to visualize this trend,
this figure, figure 3,
plus the mean AI over the 200 communication steps for both human and computer agents in the condition.

You can clearly observe that both MH and AA are always acceptance conditions show a general trend to increase AI for both humans and computer, indicating successful information integration and learning from interaction. and in stark contrast, the AR, AR condition, always reject condition, whereas the AI always reject proposal, show a little to no improvement from the initial categorization accuracy, especially for the computer agent.

This graph vividly demonstrate the superior performance and learning trajectory achieved under the Metropolis Hastings interaction, particularly for the AI agent.

and the other is the same.

Okay,

Beyond the just categorization accuracy, we also evaluated the sine posterior distribution convergence.

As highlighted in table 1,

the MH group achieves the highest convergence between the agent's final sine distribution and the target posterior distribution derived from the combined model.

This indicates that the MH-based interaction most effectively guided the diad towards the emergence of a shared informative sign system that truly reflects the integrated information.

While human participants in the MH condition also showed the high agreement, the difference compared to the always accept condition was not statistically significant.

So, however,

the AI's performance in the MH group was significantly superior, demonstrating its ability to converge the interaction to align its understanding with the humans, leading to a richer shared representation as indicated by the convergence toward the S given XAI and Xhuman.

To summarize our discussion,

our experiments clearly showed that the human-AI pair can successfully engage in co-creative learning via the joint attention naming game.

So, under partial observation,

the Metropolis Hastings acceptance rule significantly improved categorization accuracy and agreement the same as compared to both always accept and always reject control conditions.

This is evident in the AR-I graph and the sign agreement score we just discussed.

The other way,

the human-AI's acceptance behavior is that the human-AI's behavior aligns closely with the theoretically derived

Metropolis Hastings probability as shown in this figure 4.

This figure is shown in this figure.

This reinforces the idea that the mechanism enhances perceptual integration and effectively facilitate decentralized Bayesian inference within human-AI diet.

These findings collectively suggest a promising direction of designing truly symbiotic AI systems,

systems that run with human by fostering the emergent,

and the interaction-independent,

and the interaction-driven alignment,

moving beyond the limitation of traditional unilateral approach to the AI development.

Okay, so while our findings are highly promising,

we acknowledge several limitations in current studies,

we utilized actually simple the artificial-generated visual stimuli and categorical signs.

So this control setup was necessary for initial validation, I believe,

but it naturally meets the generalizability of our findings to complex the complex real-world tasks involving richer semantics and natural language such as those handled by large language models.

Additionally, our joint attention naming game,

Paradigm assumes the perfect joint attention between agents.

So in real-world interactions,

so attentional alignment can fail,

which is an important,

important aspect for future research.

And looking ahead, we identify several key future challenges.

These include,

scaling to richer modality and semantics,

robustness to imperfect attention,

designing real-time co-adaptive system,

and understanding emerging dynamics in large-scale systems,

and also ethical and interpretability concern is also important, I think.

So, and actually,

to build a truly symbiotic human AI society,

how to integrate this kind of idea to the practical applications,

that is also important topic.

Okay,

that's all from my side.

Thank you very much for your attention.

And I look forward to further discussion.

Okay, thank you very much.

Thank you for the talk.

Awesome.

All right.

I will ask a few questions that I wrote,
and I will copy them into the chat as well.

And for anyone in the live chat,
please feel free to write questions and I'll copy them in.

All right.

So first comment and question.

The task focused on categorical discrete,
as well as quantitative aspects of situational semantics,
played via a language game,
related to sharing information about a shape in a shared world.

What kinds of AI human tasks do you think this kind of algorithm could be used for?

For example,
do you see it in the domain of the text chat or a robotic setting?

Or just in other words,
which application directions do you find promising or most able to demonstrate the differences and strengths
of the framework compared with the unilateral supervised or unsupervised learning systems?

Yes.

Thank you very much for great questions.

Actually, that I should ask,
that I should add in our explanation.

So actually,
maybe this experiment itself looks very,
I would say simple and categorical,
categorical sign formation.

And this,
but what I want to show in this talk and in this paper is the kind of the very basic validation of this approach,
co-creative learning.

So,
and this kind of this graphical model and this setting is not limited to the categorical setting.

And also that, you know, the small number, I mean, the simple visual image.

And actually,
so as I mentioned,
what I did is swapping the one agent in this kind of the naming language game.

The swapping the by,
swapped by a human participant.

So actually this kind of approach,

collective predictive coding based approach is now extending to many domains.

So let me share the,

okay,

so it's a kind of appendix slide.

So actually this approach was extended to the multiple agent.

And also the multi agent reinforcement learning.

And also the importantly,

you know,

this is also another preprint we recently shared.

So this approach can be extended to the more complex tasks.

So this is,

for example,

this,

this paper is dealing with the kind of the captioning game,

not naming game,

captioning game.

So between the two vision language model agent,

you know,

the one agent receives a natural image and describe it in natural language.

Okay.

So then they receive the another agent receive the text.

Text is just a vector,

you know,

a sequence of categorical variables.

And that can be treated in our framework in a similar way.

So actually that we showed that this kind of process improve,

I have to successfully integrate the two agents,

the knowledge of the vision language models.

So I think we are now,

we are many researches now ongoing.

So I think that we can,

I believe we can extend this idea to the large language model setting.

And also that we can say that now,

as you know,

currently in robotics field,

vision language action model,

vision language action model is gathering attention.

So maybe we can consider the,
how can we extend this idea to such kind of robotic scenario as well.

Awesome.

Um,
okay.

Next question.

And then I'll read from the chat.

Okay.

How do you see this in terms of converging to a shared stable truth versus situations of novelty,
like co-creation of art or research where there's not necessarily a preexisting ground truth,
even of a latent kind to converge to.

I'm thinking about situations where the agent models have overlapping information,
but I'm thinking about the,

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existing compromise versus move into a third space that might be different than what either agents has or knows thank you very much for your great question and actually that's very very very very uh interesting uh topic and uh interesting interesting uh discussion point so actually the mathematically speaking so if the if they don't have the shared information see I mean the you know the observation the contain does not contains shared the information so it's very difficult to uh I I expect that that it's very difficult to form the shared representations and reach agreement in some sense so and uh but uh the it's better and in but I didn't have I don't have the empirical evidence uh about that on the with some expense but I think that this is now it's very important uh step to conduct and constructive studies studies to analyze such kind of what kind of phenomena happens with uh this kind of the you know basic mechanism of symbol emergence and uh formation of

shared the representation and also now maybe that this is a kind of the somehow a bit outside of today's topic and I think that this kind of scenario can be found in the democratic discussion in our society as well so the I think that one of my students is working on the the topic about the relationship between collective predictive coding and democracy and digital democracy so the if the two agents don't have the shared experience uh it's very difficult to have to reach or to reach agreement and also that you know the the the different if they have a very different belief system the it's it's the I'll say it will become very difficult to agree so that that's such kind of thing this this kind of mechanism is somehow existing the behind the curtain of the current the social division or system yeah I think that the currently I'm very interested in the connection between the you know this kind of computational research and the simple emergence for representation learning stuff things and the social science uh topics uh topics as well thank you very much yes that makes a lot of sense I guess just speculatively if two agents are in a shared environment with a simple optima to be reached or at least a gradient

to be followed at least there's a norm to which the gradient could be followed or local or global optima could be reached but if you have agents that are in two different environments you kind of lose even the possibility of a guarantee because if it's like if two people are in two different rooms with two different temperatures there's no guarantee they would converge to the same air conditioning but if they were in the same room with a shared air conditioning at least they could yeah okay yeah go ahead I think it's very interesting to think over the kind of the how can we how

uh should we in how can we improve the design of communication and how can we how uh should we uh consider the the environment the difference of environment uh to achieve the mutual and us to you know uh to how to say to understand the mutual understanding that between uh two or more agents something like that uh sorry yeah sorry first i'll read a question from the live chat if if anyone else has a live chat question they should write it so frederick kintanar wrote in your model how do you approach the difference between the way silicon agents and humans model the concept level of words and context ai is opaque how can this be improved for interpretability okay thank you very much that's also very much important

point that in our framework the most important part of our framework is that we totally deal different to how say totally deal with the system a kind of decentralized system so the in the one of the most important part of this mechanism we that each agent need not to look into the the internal phenomena now of the counterpart agents so the even in the computer agent settings so what the agent do is just a kind of the uh utter the giving sign to speak something and listen and and determine the if it accept or not the by itself so so that there is no you know there is no need to uh kind of kind of look into the other agents variables you know so the so this you know that actually that in the similar studies in emergent communication some many studies uh how to say uh ask agent to pass the kind of the gradient of the parameter through agents you know so across agents it's impossible for human but uh but uh in this setting uh we don't need to do that so actually that in this uh system mechanism we just consider the counterpart agent as a black box so what we did in this framework is in concrete planning is just swapping the black box

by human agent and just expect the human human behavior follows the mh naming game based mechanism algorithm algorithm so actually so to that end uh to validate that so we did conducted this experiment previously and we somehow found that the human in this kind of setting human actually followed the mh algorithm to a some extent and also the in this experiment again again we checked that the human behavior is similar to the image algorithm and as a result that we can expect actually expect that the total the collective free energy can be decreased decreased through this interaction that is the scenario and regarding the explainability that is a very how to say the difficult question and we need to consider the problem and uh but uh you know the better in the in the uh how to say in the series of studies of explainable a a i so the definition of accessibility as is how say very somehow uncertain you know the uh how to say the uh how to say the uh how to say the uh how to say the uh how to say the uh the uh in the current large language model the explainability i mean the what happening inside of the neural network is a very challenging problem and uh how to say the human um i would say um in i think that the similar problem i can be found in this scenario as well but uh anyway so when we introduce this kind of idea idea to our society uh we assume that uh a a is not just a follower of or students or human so the human can be affected human decision can be affected uh by ai so maybe the of course that we need to reconsider the escalal issues and uh explainability based on such kind of assumptions uh that my answer is not perfect but yeah that's comment from my side great okay and one more

question i wrote in the comment in the chat so you mentioned different ways of communication amongst ai human ai and human human dyads and groups and i think the relevance of understanding and modeling and creating high integrity settings for these communications is clear so what might these communication strategies look like for example let's assume that we can do real-time language translation between mother tongues like japanese or english from there what kinds of games might be possible for example with that high bandwidth communication still look like how are you doing what are you up to or could there be some structured communication games like captioning or joint classification which could engage shared attention in ways that were previously impossible and discover new global free energy optima uh thank you very much for your questions i'm now have to try to grasp the point we can do real-time language translation this is um mother tongue like uh from there what kind of game if everyone could speak and hear

in the natural language that was most comfortable to them what would be the semantics of the language games

that would support high bandwidth communication high bandwidth communication

uh you mean the uh you mean the the i so what what what what does the real-time language translation means here so everyone can this this means everyone can understand so what is

you can speak you can speak in japanese and hear what i'm saying in japanese

and i'll hear what you say in english and i could speak in english so we're using some ai system to auto translate what we hear and say

in close to real time so that so we can have a natural conversation with our mother tongues

and we can speak in the language so then i'm just kind of curious and and wondering whether then we would have

a sort of normal like on the street conversation or whether your work is providing some situations or more structured games like kind of like language chess where we could play a game together like captioning or classification that could be used to reduce the free energy on a joint distribution

in a way that would be like faster or different than just asking about your day or your friends or your family

all right thank you for your question and uh now i'm not still uh i'm not sure if i got get the

uh the point clearly but uh basically the uh i know image naming game the naming game or captioning game uh this

is kind of a very very small subset of what the what's happening in our total language game the inner

society so that actually basically the our assumption is that our assumption or our idea is that uh

the language the the formation of language or symbol emergence itself is a part of the adaptation

to the environment and uh you know that's totally somehow same as uh collective uh predictive coding

and uh active influence and uh so the the you know so i i i don't have a clear answer to this

uh question uh but uh uh how to say so you know the we share some task we share some task or some environment

if you are human the that's task so maybe gradually the our way of communication uh gradually change and optimize

uh to the uh to the uh to the shared task or shared environment that that's that's my idea and i don't know

that kind of the i don't know the connection with the the real time translation uh i'll say the the discussion point so um is it okay yeah sorry i i think it's a a mystery for us

and others to explore in the future like if we were fluent in hearing and speaking to everybody what would we talk about like if we knew every language perfectly and we could communicate to to everyone and everyone could be included in the conversation would that conversation look like just meeting somebody at the bus stop or i think your work provides some structured settings where the information content of communication could be quantitatively analyzed and different contributions and proposals could be understood in terms of how they contribute to co-alignment um okay and thank you yes last question so what are your next research directions or questions and what milestones or developments from your group and colleagues or in the broader ai space are you looking

out for okay thank you very much in the relation with the development of my ai system so the one thing is that how can we

integrate this idea into llm the agent systems and vlm and the vision language action model and actually so one of the you know the the for the basically the you know the

the cpc the original idea is cpc and the symbol emergence systems so the i would like to jump out that from this kind of agent viewpoint but uh the i think that this is crucial to consider

the how can we how can we how can we form the language system or shared the belief system together with other agent that i think it's crucial for human intelligence i think and cognition i think and uh the and from this viewpoint we can i think we can integrate

the human and the human and the robot uh to this system so and that can bring us to the next step of ai human and ai the symbiotic society so that is important point and the first one step is to the make the this kind of the this kind of system work the with ai and robot i mean that you know i showed the vlm the image captioning game system so actually that actually

input improve the performance of vlm and uh also that i can i think that llm that can can be uh let's say the you know the kind of the decentralized

uh or multi-stated or multi-agent or multi-agent system can be uh reformulated from this viewpoint and also and we can put the human agent uh to such such kind of ecosystems you know so uh of course the human llm interaction uh can take time uh can take time and uh and through that process so llm can learn from the human participants the you know adaptively but at the same time so inevitably human agents can be affected by llm's output but in the in many discussions this kind of the phenomena is regarded as just a bias or bias somehow the bias is constant to be considered as a negative thing negative thing but uh you know the such kind of buyers it can be regarded as a kind of the knowledge the from ai agent as well so the we i think that the the most important one of the most important next step is to formalize this kind of interaction from the viewpoint of the collective free energy minimization and the language image like language game and you know produce the kind of qualitatively different the human ai ai the

uh system and that hopefully guarantee the co-creative process and that is the most important uh step uh to forward yes thank you very much thank you a great way to end it i really appreciated the the presentation i think that the comparison of supervised or unsupervised learning makes it clear

that both of the unilateral frameworks have fundamental limitations and that a co-creative path
is a great joint distributional future thank you very much thank you okay till next time
see you thank you very much
um
Thank you.